



West Bengal State Electricity Distribution Co. Ltd.

(A Govt. of West Bengal Enterprise)

Vidyut Bhawan, Bidhannagar,
Kolkata - 700 091

Circular No: 05/2011

Dated: 01.03.2011

Sub: Design, quality and other parameters of PPE for observing safety norms

It is being observed that site officer in-charges can hardly take decision in assessment of requisite numbers of essential PPEs (Personal Protective Equipments) for technical workers/employees. In order to set a standard pattern and also to reduce wastage of time for searching different alternatives by field functionaries, the specification of PPE's are set out below.

To ensure availability & use of sufficient numbers of PPEs, following yardstick has been furnished to be followed scrupulously by the site in-charges. This yard stick is equally applicable for enlisted erection contractors/SHGs etc. All controlling officers of the Distribution (O&M) Division, Project Managers (RE), in-charges of Hydel units and PPSP etc. are requested to act accordingly.

Fire Extinguisher for 33/11 K.V. Sub-stn.:

33KV sub-stations are prone to fire hazard like electrical fire from electrical installations, fire from flammable oil of transformer, O.C.B etc and also fire of solid combustible materials. Considering the above type of fire hazard, the following types of fire extinguishers along with sand bucket suitable to combat fire at 33KV sub-station are suggested.

Sl. No	Type of Fire Extinguisher	Portable type (Capacity)	Required No.
1	Carbon dioxide (CO) type (BC) I.S:2878	Preferable 4.5 Kg	02
2	Dry Chemical Powder(BC)/ABC Dry Powder type. I.S:-171/ I.S:13849	Preferable- 5 kg	02
3.	Mechanical Foam type (AB) I.S:10204	9 liter.	01
4.	Water (CO)Type (A) I.S:940	9 liter.	01 To be kept at left side of the main entrance of the control room.
5.	Sand bucket with Stand I.S:2546	9 liter	4 Buckets are to be kept under shed by the side of each entrance gate coming from Switch Yard to Control Room.

Contd.....P/2

- Note: 1) Fire Extinguisher shall be placed on supporting wall or in cabinets in such a way that their bottom should be 750 mm to 1000 mm above the floor level.
- 2) All extinguishers to be re-filled immediately after discharge.
- 3) All firefighting equipment shall be subjected to routine maintenance, inspection and testing with record.
- 4) All employees/ employees of Agency should be convergent with the operation & use of Fire Extinguisher.
- 5) Item under sl . no. 1, 2, & 3 are to be kept hanging at easy accessible wall in front of the switch gear & panel.
- 6) Sl. no. 3. & 4. are to be used at "Power off" position.
- 7) Operation of all such kits be made by departmental employees and by the contractor's employees wherever it is out sourced. A training shall be organized by the Divisional Manager concerned.

Fire Extinguisher for Gr. E/S. or Other Similar Offices

Group Electric Supplies are exposed to fire hazard like electrical fire from electrical wiring, fire involving electrical equipments and delicate equipments, i.e., Computer and its accessories, A.C machine, Xerox machine etc and also fire from solid combustible materials. Considering the above type of fire hazard, the following types of fire extinguishers are suggested at Gr. E/S to combat fire.

Sl. No	Type of Fire Extinguisher	Portable type (Capacity)	Required no
1	Carbon dioxide (CO) type I.S:2878 *Most suitable for "B","C" type & "Electrical" fire.	Preferable 4.5 Kg.	03 nos. per 1000 sq ft. floor area *The extinguishers shall be placed in conspicuous locations over the entire floor area especially at cash, collection kiosk, computer etc. and shall be readily accessible for immediate use at the time of emergency.
2.	Water (CO)Type I.S:940 * Not suitable for "electrical" fire until make it dead/de-energized from source. Most suitable for "A" type fire.	9 liter	Total 02 nos. in a Gr. Supply *To be kept at any side of the main entrance to the supply office under covered & protected area and shall be readily available at the time of emergency.

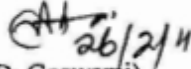
- Note: 1) Fire Extinguisher shall be mounted on supporting wall or in cabinets in such a way that their bottom should be 750 mm to 1000 mm above the floor level.
- 2) All extinguishers to be re-filled immediately after discharge.
- 3) All firefighting equipment shall be subjected to routine maintenance, inspection and testing with record which shall be carried out by properly trained person.
- 4) All employees/ employees of agency should be conversant with the operation & use of Fire Extinguisher.

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Personal Protective Equipments for Technical Staff to be used during work

For various activities of work at Distribution Network

Sl. No.	Name of the Equipment	Specification	Quantity to be issued to	Remarks
1.	Safety Belt	Full body harness with Waist Belt and accessories IS-3521	60% of total Line-staff/gang.	Till serviceable
2.	Discharge cum Earthing Rod	Highly insulated FRP Telescopic Type, with all accessories suitable for 11 – 33 KV system, 100 KV, ERDA type tested.	2 (two) nos./ gang.	Till serviceable
3.	Safety Helmet	Made of polyethylene/glass fiber plastic reinforced with strap fittings IS-2925	100% of total tech. staff/gang	Till serviceable
4.	Hand gloves (pair)	Made of rubber for electrical work, suitable for 11 KV system, IS-4770	100% of total tech. staff	Till serviceable
5.	Gum Boot (pair)	Made of rubber	30% of total tech. staff. Operational Staff posted at 33/11 KV S/Stns. must be provided.	To be issued once in every four years.
6.	First-Aid box	One bottle-Potassium permanganate crystal, 12 nos. roller & triangular bandage, Sterilized cotton wool, one Bottle mercurochrome(2%) solution in water, aspirin tablet, Burn gel with burn-dressing, scissors, Dettol	One set per gang	Each items to be issued as per requirement
7.	Bamboo ladder	Made of suitable matured seasoned bamboo, steps of ladder should be guarded by G.I. wire binding	One number for each gang	As and when required
8.	Lifting Rope	Preferably high quality nylon rope	6 bundles for each gang	As per requirement
9.	Earth spike (G.I.) (Required for discharge when existing earthing is not proper)	As per standard specification	Two numbers for each gang	As per requirement


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